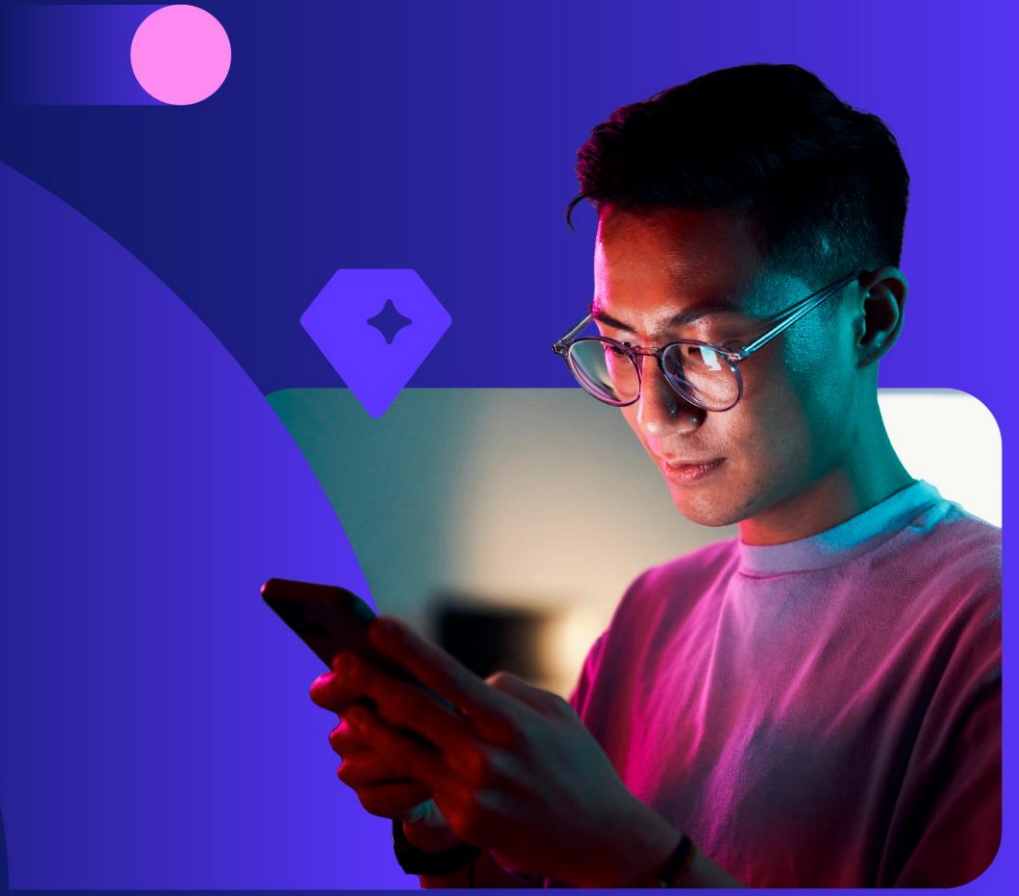
SAP TechEd

TM268

Analyze and optimize ABAP Cloud model apps built on core data services

Robert Werlich, SAP
Steffen Siegmund, SAP
Ingo Braeuninger, SAP
Thomas Fiedler, SAP

October 2026



SAP TechEd

Agenda

Today’s scenario – Overview

Today’s scenario – Exercises

Wrap up



Today’s scenario: Overview

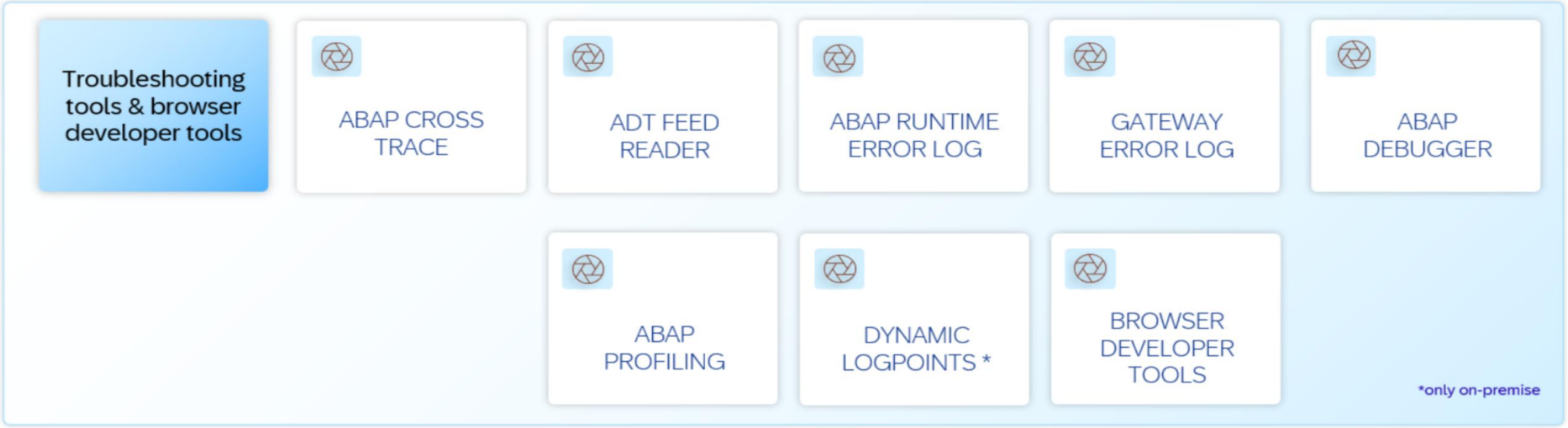
Troubleshooting tools for ABAP Cloud applications

ABAP CROSS TRACE
Designed for RAP-related ABAP functionalities
Provides insight into the RAP runtime framework, can trace data (like a payload trace in SAP Gateway).

SQL ANALYZER FOR SAP HANA
Designed for analyzing SQL query performance on SAP HANA. Provides insight into the execution plan chosen by the HANA optimizer, can identify bottlenecks like full table scans, missing indexes, or inefficient joins

BROWSER DEVELOPER TOOLS
Designed for frontend and HTTP-level debugging directly in the browser. Provides insight into network traffic, OData request/response payloads, and rendering behavior, can trace API calls similarly to a payload trace in SAP Gateway. The F12 Developer Tools Console reports for SAP Fiori / UI5 applications OData binding errors, metadata resolution issues, and resource loading warnings.

Link: https://pages.community.sap.com/topics/abap-testing-analysis/troubleshooting



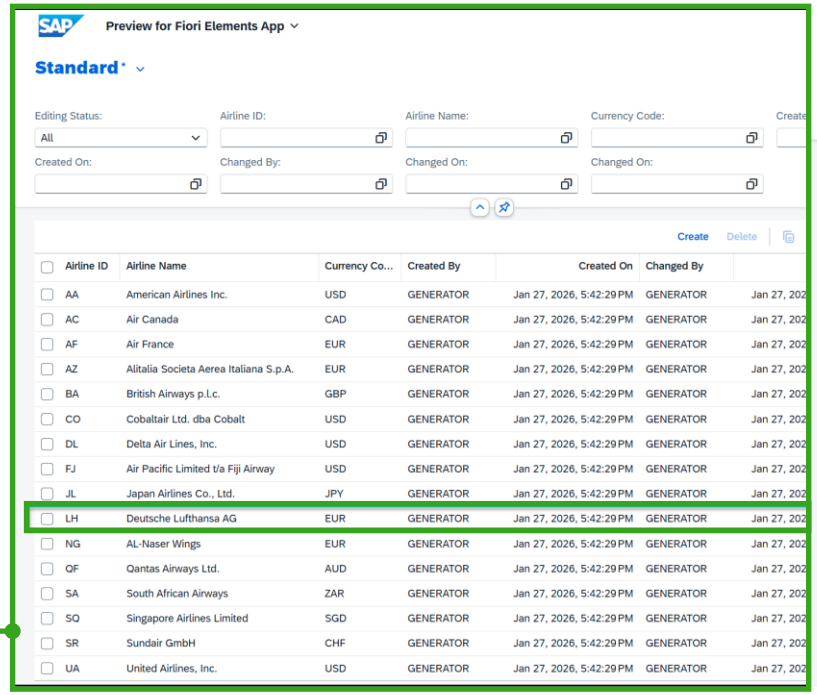
Today's scenario: RAP Application Extended with Booking Class Insights

We extend a generated RAP Application with Navigation to a Custom Read-Only UI Service

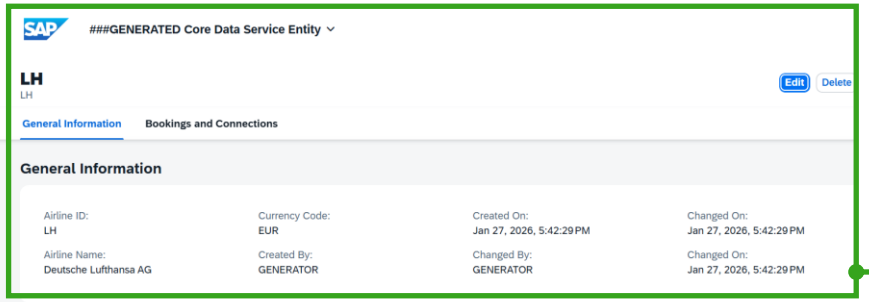
... the custom extension exposes in a RAP Application as navigation a custom CDS View (*ZTM268_C_BOOKING_CONN_CARR_###*)

... providing for price classes (Standard, Business, Premium) the number of bookings per flight connection and destination for each carrier

... for a RAP Application directly generated from database table */DMO/CARRIER*



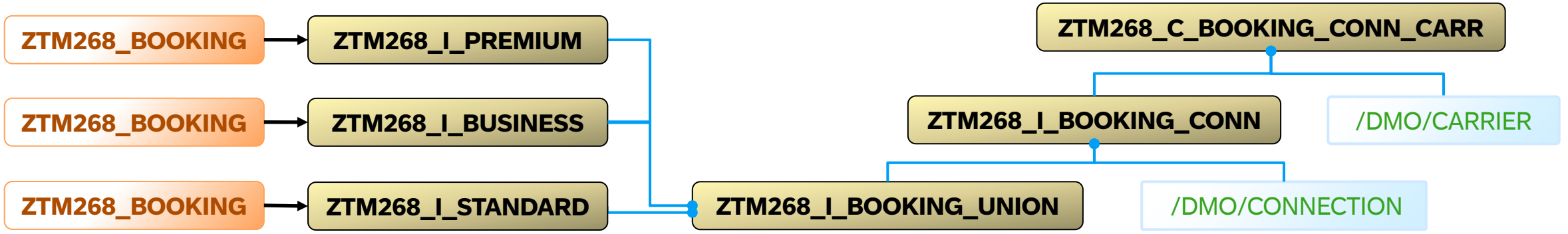
GREEN
RAP app directly generated from a database table



GREEN
RAP app directly generated from a database table

RED
Custom Extension exposing Bookings per Price Class

Today's scenario: RAP Application Extended with Booking Class Insights



STANDARD FLIGHT MODEL

Flight Price & Bookings (/DMO/BOOKING)

↑

Connection (/DMO/CONNECTION)

↑

Carrier ID (/DMO/CARRIER)

In extended RAP Application

ZTM268_BOOKING

ZTM268_CARR

ZTM268_BOOKING

CARRIER_ID	COUNTER
AZ	4.720.944
LH	1.844
JL	3.824.098
SQ	6.561.312
UA	3.307.328
AA	5.981.196

STANDARD TABLES

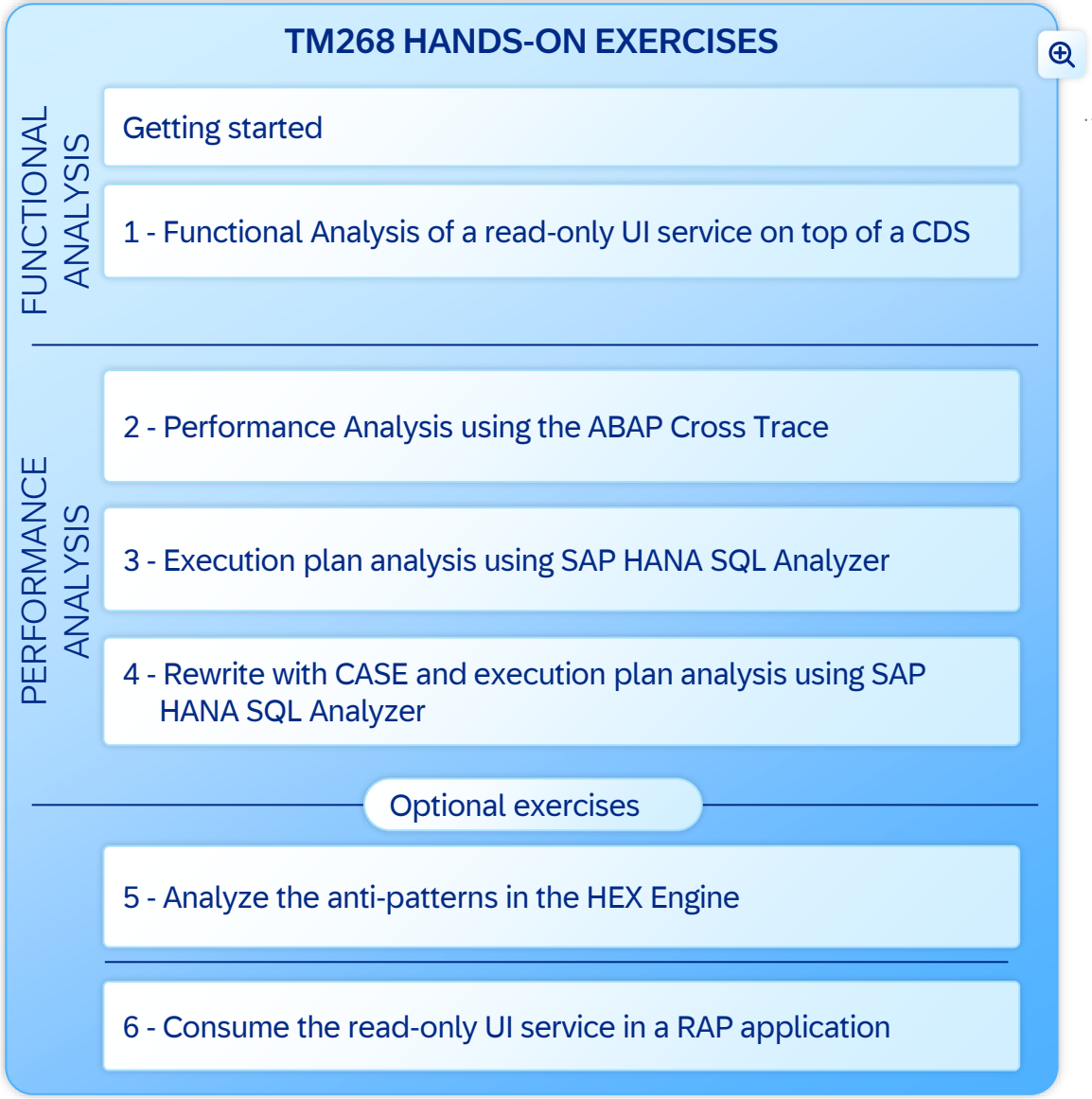
EXTENDED TABLES

CDS VIEWS



Today’s scenario: Exercises

TM268 – Analyze and optimize ABAP Cloud model apps built on CDS



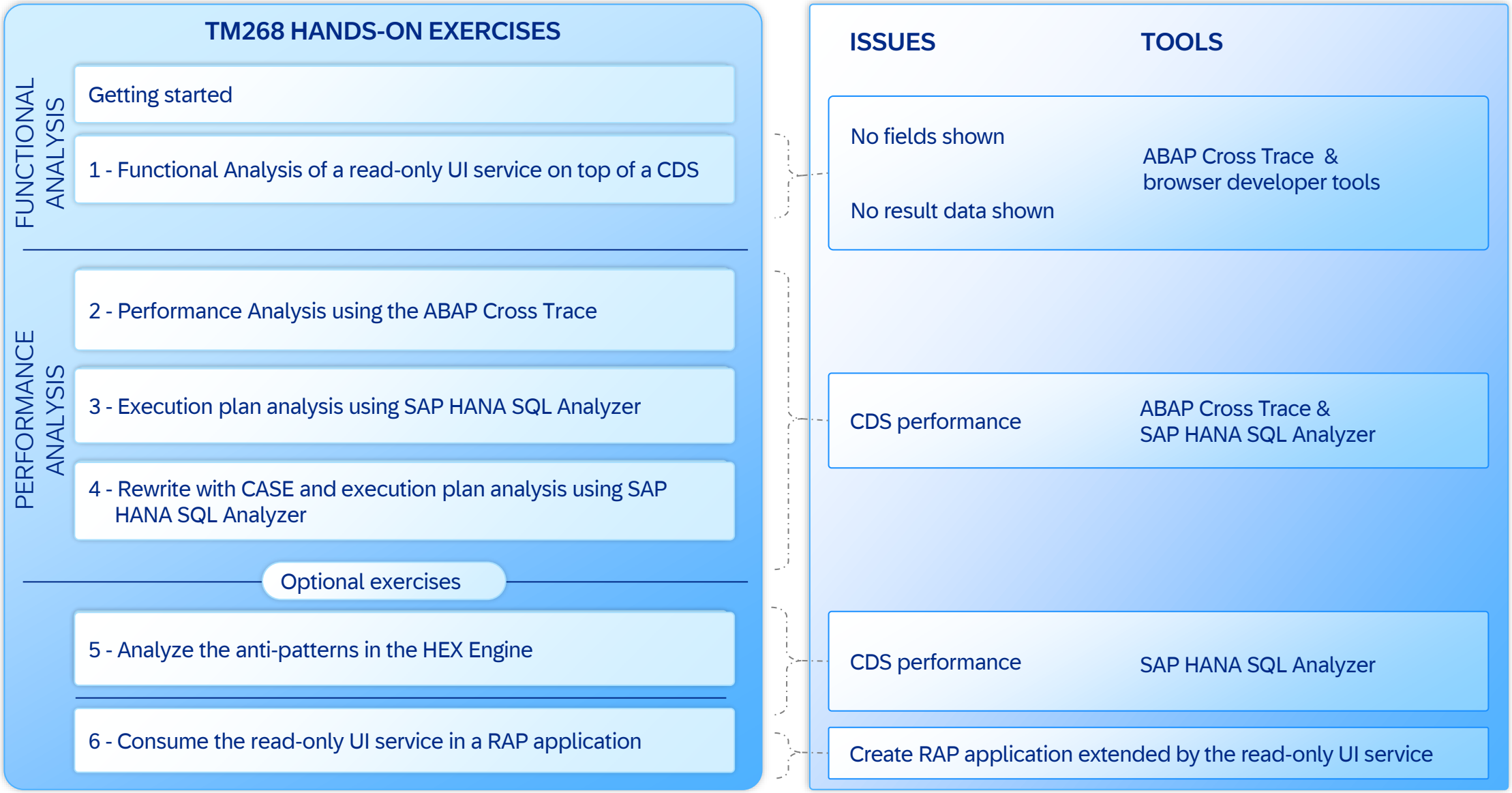
PREREQUISITES

- ✅ Set up your ABAP Development Tools for Eclipse
- ✅ Access to an appropriate ABAP system: (here: APB, see handout)

ACCESS THE EXERCISES

https://github.com/SAP-samples/teched2026-TM268 (Link)

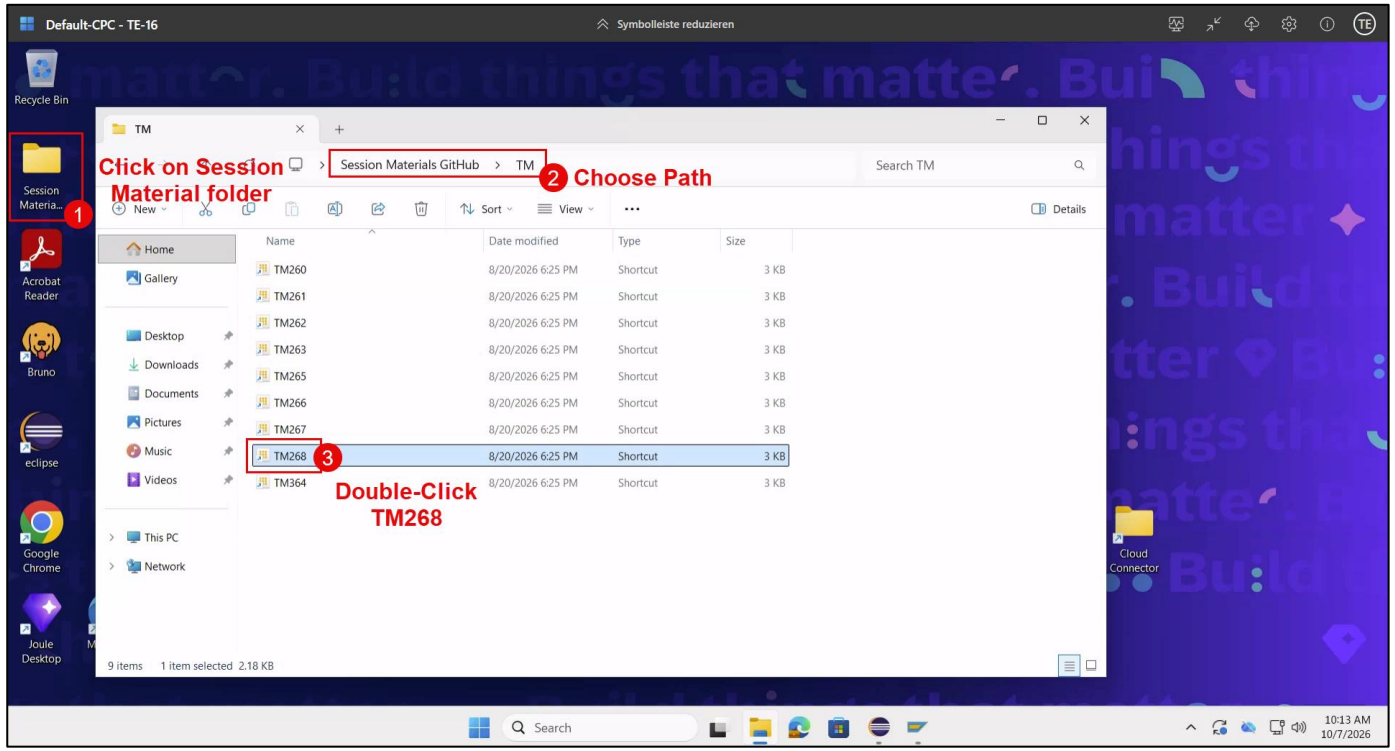
TM268 – Analyze and optimize ABAP Cloud model apps built on CDS

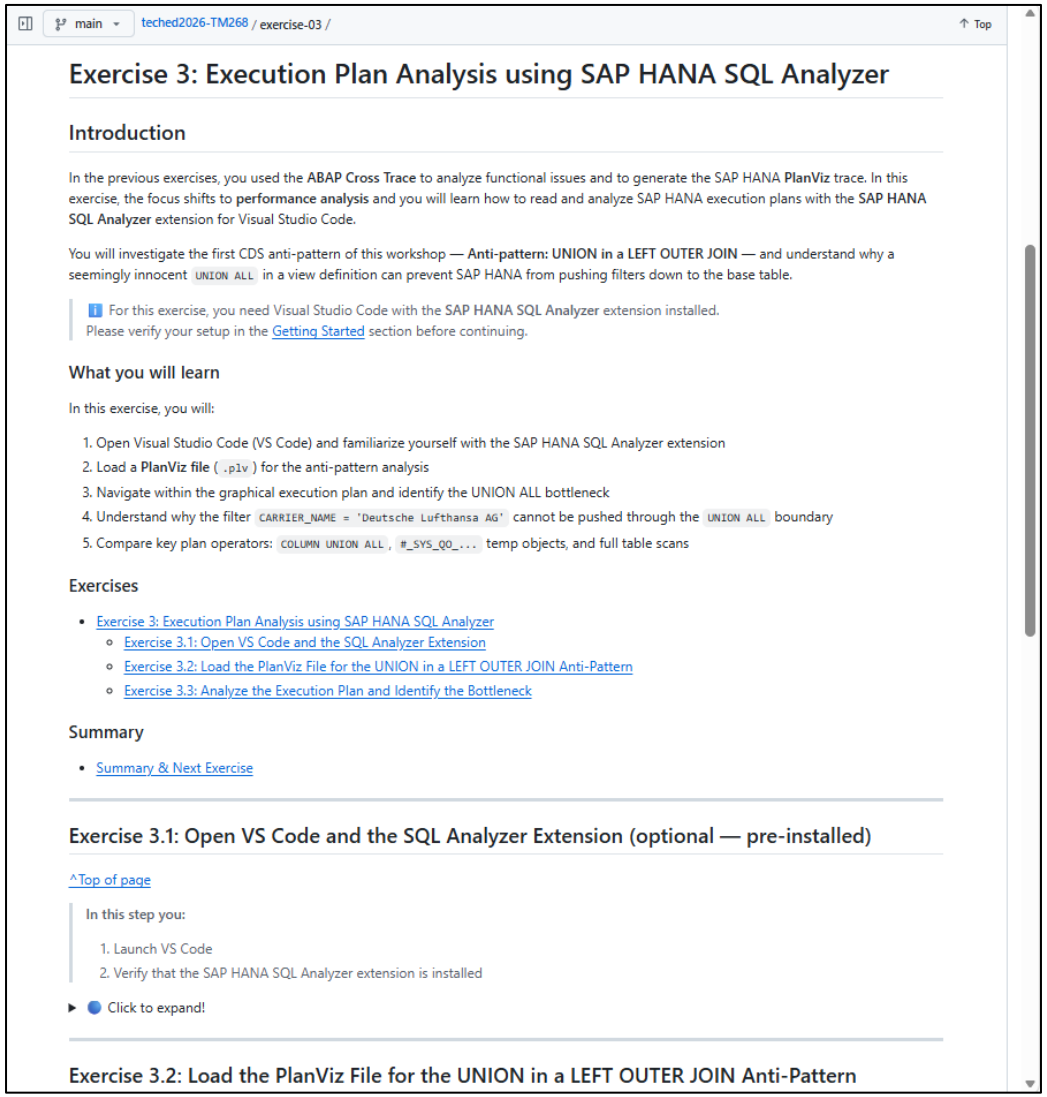


* In Visual Studio Code

The TM268 hands-on exercises – Working on GitHub (1)

Follow the step-by-step instructions provided on GitHub





main teched2026-TM268 / exercise-03 / Top

Exercise 3: Execution Plan Analysis using SAP HANA SQL Analyzer

Introduction

In the previous exercises, you used the **ABAP Cross Trace** to analyze functional issues and to generate the SAP HANA **PlanViz** trace. In this exercise, the focus shifts to **performance analysis** and you will learn how to read and analyze SAP HANA execution plans with the **SAP HANA SQL Analyzer** extension for Visual Studio Code.

You will investigate the first CDS anti-pattern of this workshop — **Anti-pattern: UNION in a LEFT OUTER JOIN** — and understand why a seemingly innocent `UNION ALL` in a view definition can prevent SAP HANA from pushing filters down to the base table.

> For this exercise, you need Visual Studio Code with the **SAP HANA SQL Analyzer** extension installed.
> Please verify your setup in the Getting Started section before continuing.

What you will learn

In this exercise, you will:

1. Open Visual Studio Code (VS Code) and familiarize yourself with the SAP HANA SQL Analyzer extension
2. Load a **PlanViz file** (`.plv`) for the anti-pattern analysis
3. Navigate within the graphical execution plan and identify the UNION ALL bottleneck
4. Understand why the filter `CARRIER_NAME = 'Deutsche Lufthansa AG'` cannot be pushed through the `UNION ALL` boundary
5. Compare key plan operators: `COLUMN UNION ALL` , `#_SYS_QO_...` temp objects, and full table scans

Exercises

- Exercise 3: Execution Plan Analysis using SAP HANA SQL Analyzer
  - Exercise 3.1: Open VS Code and the SQL Analyzer Extension
  - Exercise 3.2: Load the PlanViz File for the UNION in a LEFT OUTER JOIN Anti-Pattern
  - Exercise 3.3: Analyze the Execution Plan and Identify the Bottleneck

Summary

- Summary & Next Exercise

Exercise 3.1: Open VS Code and the SQL Analyzer Extension (optional — pre-installed)

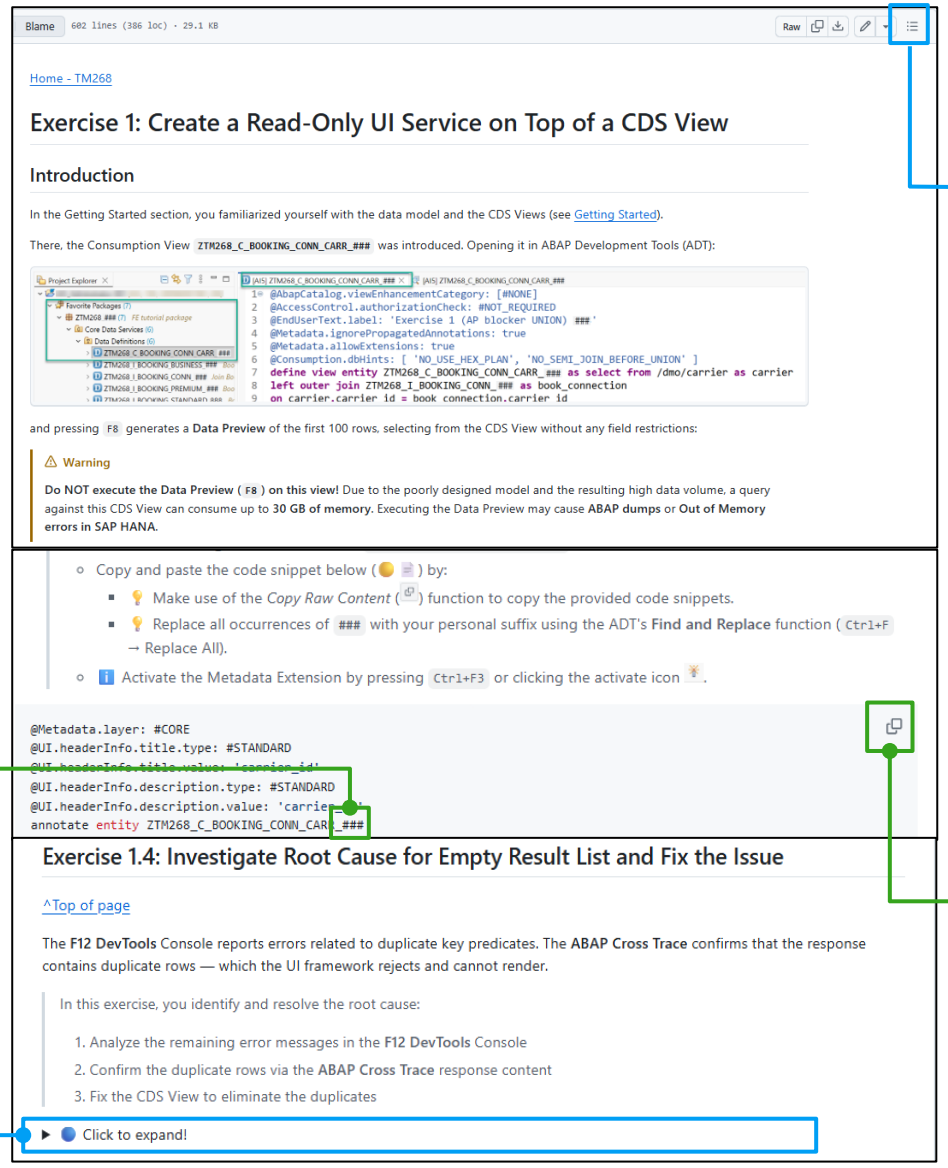
^Top of page

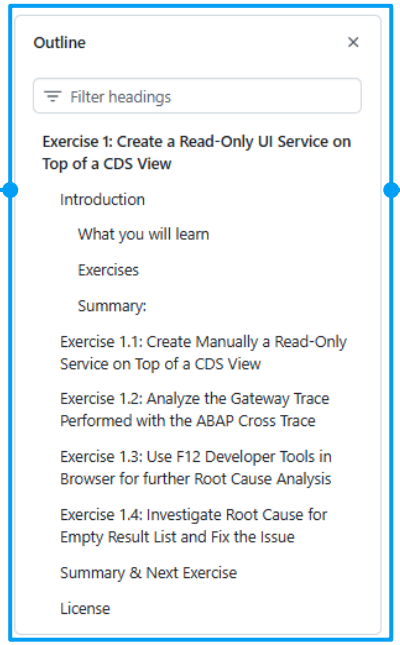
> In this step you:
>
> 1. Launch VS Code
> 2. Verify that the SAP HANA SQL Analyzer extension is installed

▶ Click to expand!

Exercise 3.2: Load the PlanViz File for the UNION in a LEFT OUTER JOIN Anti-Pattern

The TM268 hands-on exercises – Working on GitHub (2)





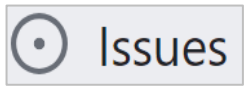
Use outline for a **quick overview**

Replace all occurrences of ### with your suffix

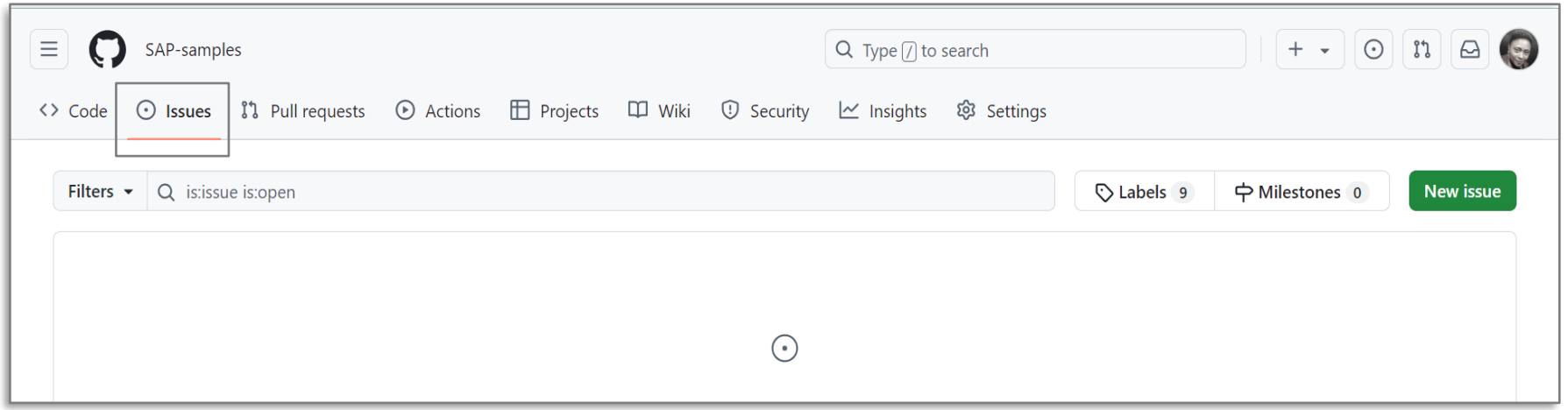
Copy raw content

Expand to see details

The TM268 hands-on exercises – Working on GitHub (3)

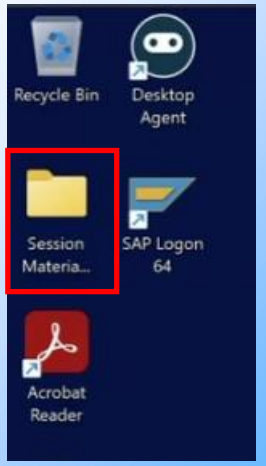


Open GitHub Issues to get help from the SAP experts

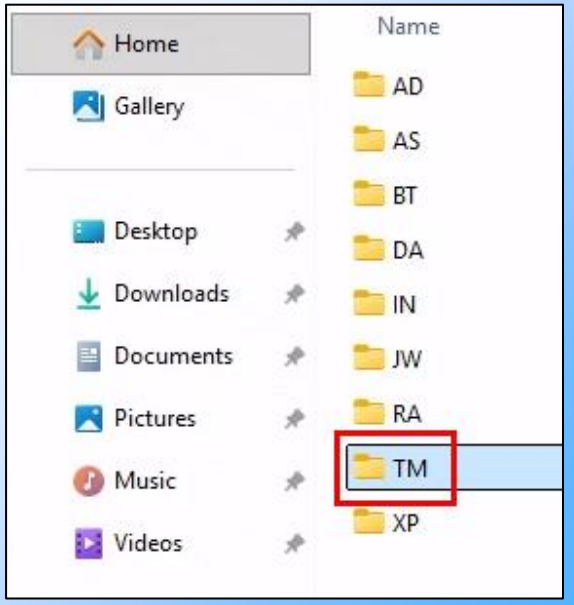


How to access the exercise material

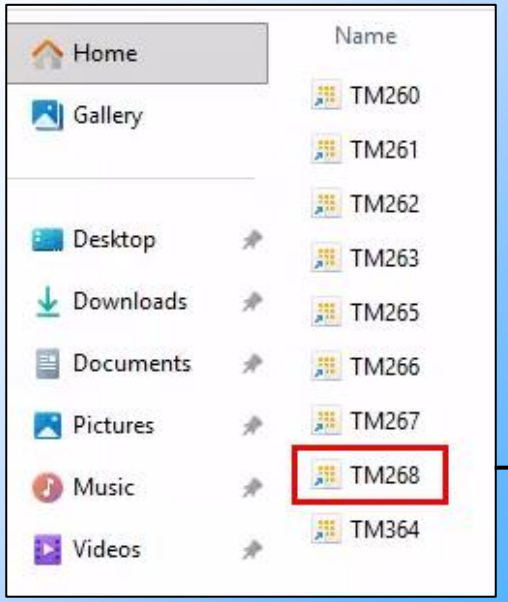
(1) Open the folder **Session Materials GitHub** on the desktop



(2) Open the folder **TM**



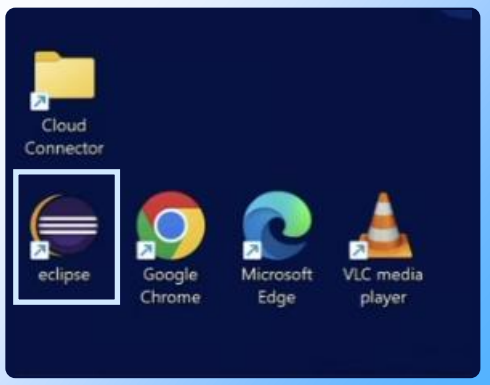
(3) Double-click the shortcut Open the folder **TM268** to navigate to the hands-on exercise



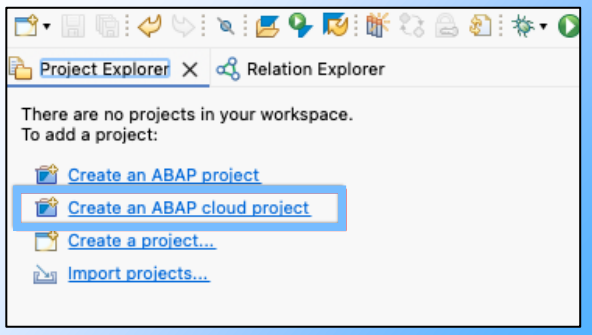
→ **TM268**

How to access your package in Eclipse

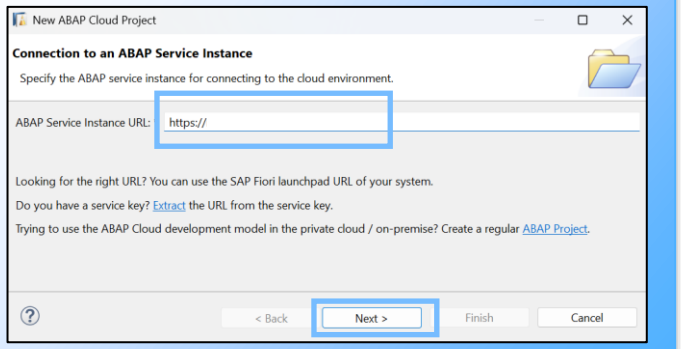
(1) Double-click the eclipse icon on the desktop to start it



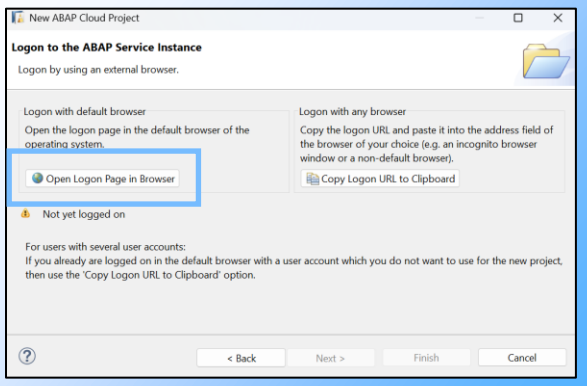
(2) Create a new ABAP Cloud Project



(3) Enter the URL of system and click next



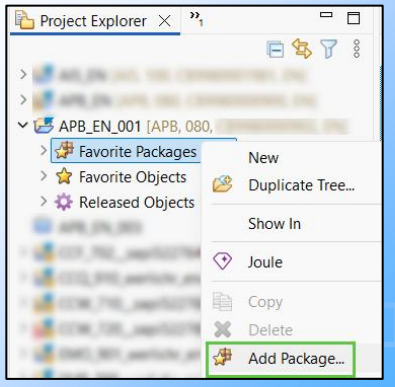
(4) Click on open logon page in browser



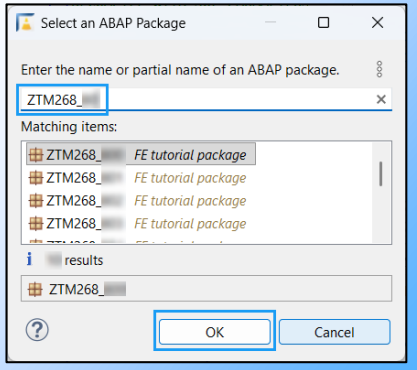
(5) Enter user name and password



(6) Add your package to favorite packages



(7) Enter with your group number ###



LET'S GET STARTED!



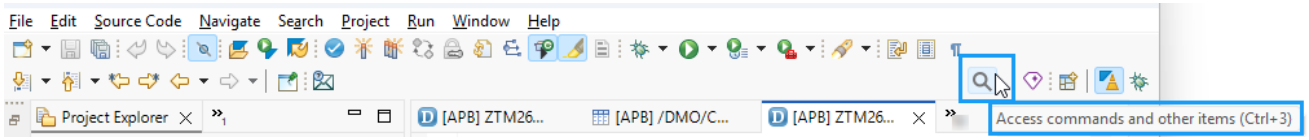
Move at your own pace!

Feel free to ask questions!

ACCESS THE EXERCISES

https://github.com/SAP-samples/teched2026-TM268 (Link)

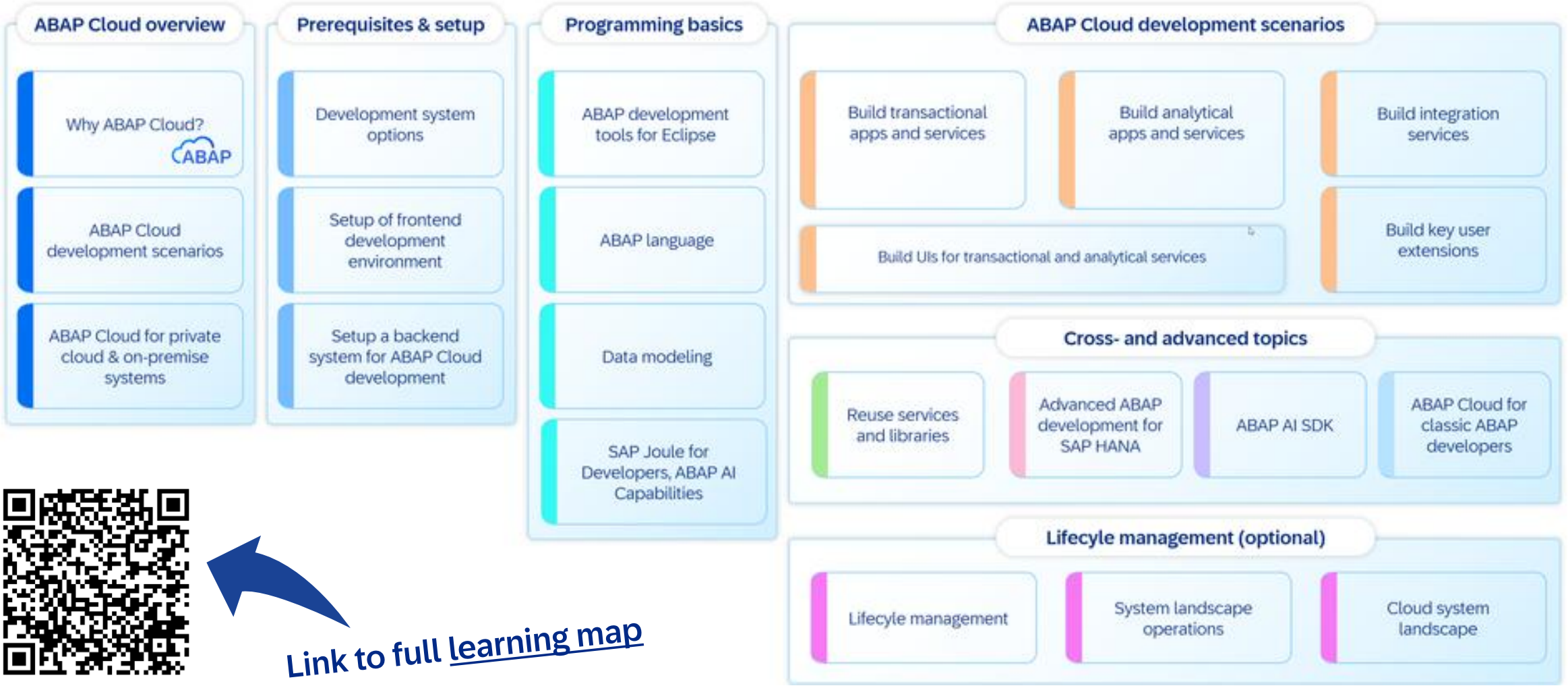
ACCESS THE TOOLS





Wrap up

Get started with the ABAP Cloud learning map



Link to full learning map

What do I do now?

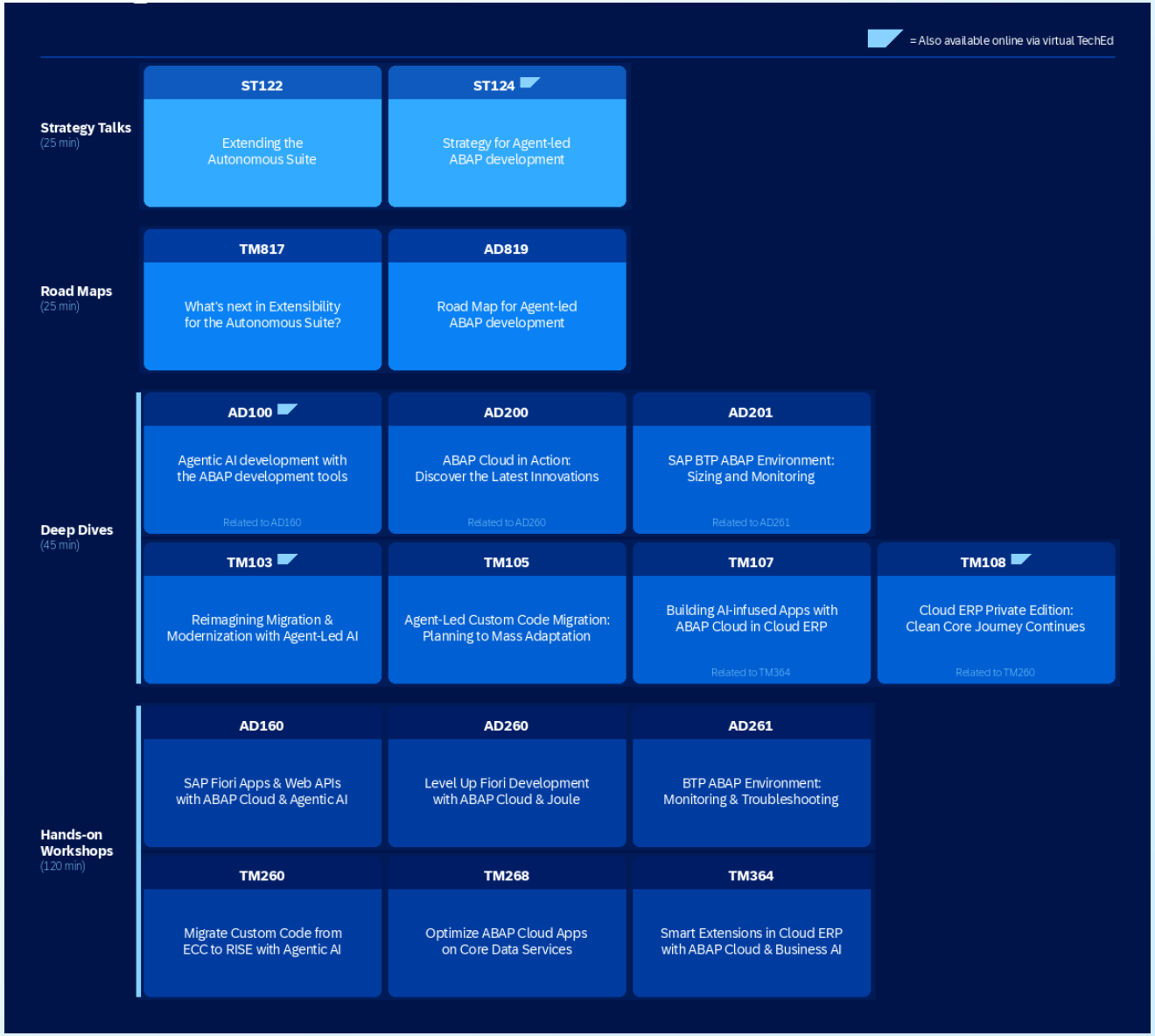
LEARN & EXPLORE

Get certified as ABAP Cloud Back-End Developer (Link)



ABAP Cloud–related sessions at SAP TechEd Berlin 2026

EXPLORE THE SESSION CATALOG (Link)



Visit us at the Demo Stations on the show floor!

Thank you.

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